



Mechanical and microstructural assessment of copper slag, rice husk ash and alkali activator composite as a sustainable pavement construction material: experimental and sustainability approach

Kuldeep Sharma¹ · Kunal Jee Thakur² · Arvind Kumar² · Sarat Kumar Das¹

Received: 9 March 2025 / Accepted: 16 April 2026

© The Author(s), under exclusive licence to Springer Nature Japan KK, part of Springer Nature 2026

Abstract

This study investigates the potential use of copper slag (CS) and rice husk ash (RHA) in pavement construction through geopolymerization. CS and RHA were alkali activated with sodium hydroxide (NaOH) (SH) and sodium silicate (Na₂SiO₃) (SS) at a 10 M concentration. A fixed SH-to-SS ratio of 1:2 was maintained. The RHA and CS varying from 10 to 50% and 90–50%, respectively, and alkali activator (AA) contents ranging from 3 to 15%. Standard Proctor compaction, unconfined compressive strength, Split tensile strength, FE-SEM, and XRD analyses were conducted to evaluate it as a pavement material. The increasing RHA decreases MDD while increasing OMC, whereas higher AA content reduces OMC across all mixes. The maximum dry density (17.35 kN/m³) was observed at 10% RHA, 90% CS, and 3% AA. The unconfined compressive strength (UCS) increased up to 20% RHA and then decreased beyond 30–50%, with the highest UCS of 10,234.47 kPa recorded at 20% RHA and 15% AA after 28 days of curing. The CS-RHA-AA composite mixes are found suitable for sub-base and subgrade layers of flexible pavements based on above properties. Multilinear regression analysis, using 80% of the data for training and 20% for testing are used to develop correlations. Furthermore, the environmental impact assessment revealed that alkali-activated mixes exhibited a significantly lower environmental footprint compared to cement-based composites. Overall, CS and RHA blended with AA offer a sustainable alternative pavement construction material, contributing to CO₂ reduction and effective waste utilization.

Keywords Geopolymer · Rice husk ash · Copper slag · UCS · STS · LCA · Microstructure

Introduction

The rapid pace of industrialization and infrastructure development has led to a significant rise in global industrial waste generation, creating major environmental and sustainability challenges [1, 2]. Improper management and disposal of industrial by-products contribute to soil and water contamination, resource depletion, and greenhouse gas emissions [1, 3]. Conventional disposal practices, such as landfilling and open dumping, consume valuable land resources and

pose long-term ecological risks [4]. In response, the valorization of industrial residues as construction materials has gained worldwide attention as a key strategy within the circular economy framework [5]. Since the construction sector is one of the largest consumers of natural raw materials, it offers an ideal pathway for converting industrial by-products into sustainable infrastructure materials, reducing dependence on virgin resources and mitigating environmental impacts [6, 7].

With mass connectivity, there is a huge demand for cost effective and environmentally sustainable construction materials. Conventional stabilization of pavement subgrades and base layers typically relies on cement and lime binders to improve load-bearing capacity and stiffness [4]. This high environmental burden of cement based binder underscores the need for alternative, eco-efficient binders capable of achieving comparable or superior mechanical performance at lower carbon intensity. In this context, alkali-activated

✉ Kuldeep Sharma
kuldeepsha333@gmail.com

¹ Department of Civil Engineering, Indian Institute of Technology (ISM), Dhanbad, Jharkhand, India

² Department of Civil Engineering, Dr B R Ambedkar National Institute of Technology, Jalandhar, Punjab, India

materials (AAMs), often referred to as geopolymers, have emerged as next-generation binders offering exceptional mechanical strength, durability, and sustainability [8–11].

The concept of alkali activation dates back to Purdon's pioneering work (1940) on activating blast furnace slag with sodium hydroxide [12], later advanced by Davidovits (1978), who coined the term "geopolymerization". Since then, AAM research has expanded rapidly, demonstrating the versatility of these binders in producing high-strength, chemically stable, and durable composites from various industrial and agricultural residues [13]. Compared to Ordinary Portland Cement (OPC), alkali-activated binders can reduce carbon emission by up to 75–80% while providing superior resistance to chemical and thermal degradation [14–16]. AAMs are synthesized through chemical reactions between aluminosilicate-rich precursors (such as fly ash, slag, RHA, metakaolin or other industrial residues) and alkaline activators like SH or SS (Na_2SiO_3) [13–15, 17]. This reaction forms a three-dimensional polymeric network composed of sodium or calcium aluminosilicate hydrate gels (N-A-S-H and C-A-S-H), which are responsible for strength and binding properties [7, 18–20]. The concentration and ratio of these activators critically influence reaction kinetics, polymerization degree, and the resulting mechanical performance of the geopolymer matrix [21, 22]. Geopolymer technology supports waste valorization and resource conservation while delivering excellent mechanical and durability properties [16, 23].

Only a limited number of studies have explored the use of alkali activators (AA) for pavement subgrade applications, and most of these are restricted to small activator dosages. Moreover, previous research has primarily focused on evaluating mechanical parameters such as UCS, CBR, and STS, with limited attention to comprehensive microstructural or environmental analyses. To date, no study has specifically investigated the potential of a CS-RHA-AA composite as a sustainable material for pavement subgrade stabilization.

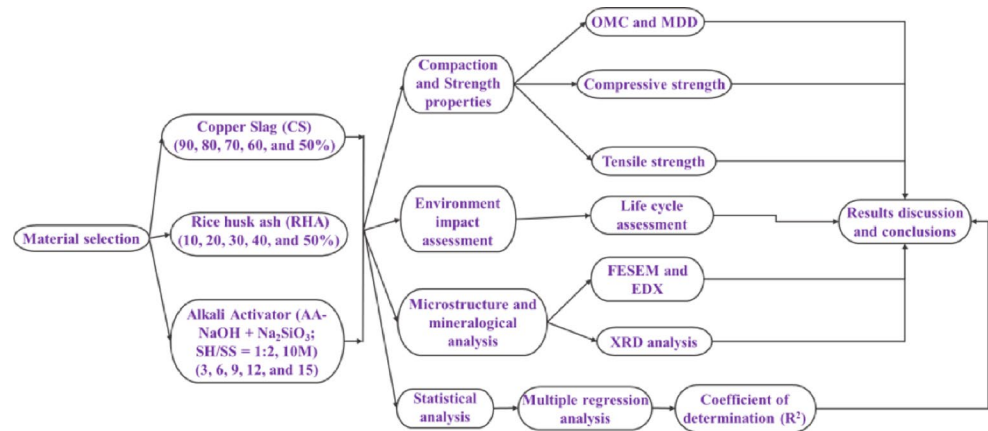
Among the various industrial wastes, copper slag (CS) and rice husk ash (RHA) are particularly promising due to their abundance and favourable physicochemical properties [24]. CS, a by-product of the copper smelting process, is produced at an annual rate exceeding 37.7 million tonnes globally [25, 26], typically in quantities two to three times greater than copper output [27]. Chemically, CS contains Fe_2O_3 (35–45%), SiO_2 (35–40%), Al_2O_3 (10–15%), CaO (5–10%), and Cu (1–2%), with a specific gravity between 3.3 and 3.8 [25]. Its dense, vitreous, and angular particle structure makes it a suitable alternative to natural fine aggregates in concrete, embankments, and pavement materials. CS also exhibits latent pozzolanic activity [28], allowing it to react with calcium hydroxide or alkaline activators to form cementitious gels via geopolymerization [29].

Numerous studies report that partial replacement of cement or sand with CS improves mechanical strength, reduces permeability and shrinkage, and lowers overall carbon footprint [28, 30, 31].

Similarly, RHA, an agro-industrial by-product obtained from the controlled combustion of rice husks, exhibits immense potential as a reactive pozzolanic or geopolymeric precursor [32]. Approximately 200 kg of husk and 40 kg of ash are produced for every tonne of rice processed [4, 33, 34]. RHA contains 85–95% amorphous silica and is classified as a reactive pozzolanic material under ASTM C168 [35–37]. Its reactivity depends on combustion temperature, with temperatures between 500 °C and 700 °C yielding the most amorphous and reactive silica [38]. When incorporated into soil or binder systems, RHA improves compressive strength, shear resistance, and dimensional stability while reducing plasticity and shrinkage [39, 40]. Substituting cement or lime with RHA also significantly reduces CO_2 emissions, making it a key component in low-carbon construction [41, 42].

Based on the literature, CS-RHA-AA composites have not yet been comprehensively evaluated as potential pavement construction materials. The strength of stabilized materials is commonly evaluated through compressive testing, while their tensile strength remains less explored. Understanding tensile strength is vital for assessing the stability of structures such as earth dams, highways, and airport pavements. As pavements endure traffic loads, soil shrinkage, temperature variations, and wetting-drying cycles, tensile stresses develop, impacting stability.

In the present study, a series of tests were conducted to assess the feasibility of utilizing CS and RHA, activated with AA, as a binder for pavement subgrade. The CS, an acidic industrial by-product, and RHA, an alkaline by-product, both pose environmental concerns such as soil and groundwater contamination, and ecological degradation are used to prepare a composite mix through alkali activation. Composite mixes were prepared at varying proportions to evaluate compaction characteristics, UCS, and STS, thereby assessing their mechanical performance. In addition to mechanical testing, microstructural analyses were performed on samples exhibiting superior strength characteristics. XRD was employed to identify mineralogical transformations and chemical phase development responsible for enhanced mechanical performance, while field emission scanning electron microscopy coupled with energy-dispersive X-ray spectroscopy (FE-SEM/EDX) was used to examine the microstructural evolution with curing. Furthermore, a regression model was developed based on the experimental results. Finally, a life cycle assessment (LCA) was carried out to compare the environmental impact of CS-RHA-AA composites with those of CS-RHA-cement-based materials.

Fig. 1 Experimental methodology**Table 1** Physical properties of CS and RHA

Properties	Materials	
	CS	RHA
Particle shape	Irregular	Irregular
Appearance	Black and glassy	Dark grey
Specific gravity	3.5	2.06
Fineness modulus	2.5	1.43
OMC (%)	18	64
MDD (gm/cc)	1.72	0.52
CoU (Cu)	1.54	7.08
CoC (Cc)	0.960	0.44
Particle Size (%)		
>425 micron	-	-
>300 micron	4.16	0.32
>150 micron	91.07	47.35
>75 micron	3.65	5.760
<75 micron	1.12	46.34

Table 2 Major Chemical content of CS and RHA

Properties	Materials	
	CS (%)	RHA (%)
Fe ₂ O ₃	40.77	1.70
SiO ₂	35.22	91.90
CaO	6.79	0.688
Al ₂ O ₃	2.80	1.38
MgO	1.75	0.688
K ₂ O	1.06	1.35
Na ₂ O	0.50	0.306

Table 3 Alkali activator properties

Properties	Alkali activator	
	SH	SS
Chemical formula	NaOH	Na ₂ SiO ₃
pH	14	14
Specific gravity	1.31	1.54
Percentage of Na ₂ O	-	8.0
Percentage of SiO ₂	-	26.5
Percentage of H ₂ O	-	65.5

Experimental investigation

Materials used

The source and the physical and chemical properties of CS, RHA and alkali activators are discussed in the section. The physical and chemical properties of the materials are discussed in the following sections. The details of the experimental methodology are shown in Fig. 1.

Copper slag and rice husk ash

CS was procured from Hindustan Copper Limited, Bhauruch, Gujarat, India. It has a light black color. RHA was collected from Jackpot Rice Mill, Jalandhar Punjab, India, which is black in color. The physicochemical properties are presented in Tables 1 and 2, which are consistent with those reported in previous research by Sharma and Kumar [4].

Alkali Activator

Sodium hydroxide (SH) and sodium silicate (SS) were used as alkali activators, both procured from Garg Chemical Industries Pvt. Ltd., Jalandhar, Punjab, India. SH was supplied in flake form, whereas SS was provided as a liquid with a purity of 98% and an activator modulus (Ms) of 3.31, defined as the ratio of SiO₂ to Na₂O. The SS contained 26.5% SiO₂, 8.0% Na₂O, and 65.5% H₂O, with a specific gravity of 1.54, while SH had a specific gravity of 1.31. The properties of the alkali activators are summarized in Table 3. AA was prepared by mixing SH and SS solutions in a 1:2 mass ratio. A 10 M SH solution was prepared using NaOH flakes and allowed to equilibrate for 24 h before use. Based on molecular weights, 1 gm of NaOH corresponds to 0.775

g of Na_2O . Accordingly, SS contributed 17.65 gm of SiO_2 and 5.34 gm of Na_2O , while SH added 10.32 gm of Na_2O per 100 gm of total activator. This resulted in an overall activator modulus (Ms) of approximately 1.13, which lies within the recommended range of 0.80–1.60 for geopolymer binder systems [43].

Experimental program

CS and RHA were combined with AA in varying proportions to evaluate their geotechnical performance. RHA was incorporated at 10%, 20%, 30%, 40%, and 50% by weight, while AA was added at 3%, 6%, 9%, 12%, and 15% of binder weight; correspondingly, CS accounted for 90%, 80%, 70%, 60%, and 50% of the mixture. The solid material content (CS+RHA) were maintained at 100%, with AA added separately as the binder solution. Detailed mix proportions are presented in Table 4.

The mix design for the CS-RHA-AA composites was developed by regulating the $\text{SiO}_2/\text{Al}_2\text{O}_3$ and $\text{Na}_2\text{O}/\text{Al}_2\text{O}_3$ molar ratios, the SH: SS mass ratio, and the liquid-to-solid (L/S) ratio. Based on the oxide compositions of CS

and RHA, the calculated Si/Al molar ratios increased from 26.11 to 51.61 as RHA content increased from 10% to 50%, while the Na/Al molar ratios ranged from 0.297 to 0.317 across the same replacement levels. These molar ratios ensured chemical consistency in the geopolymer precursor system and provided adequate aluminosilicate reactivity for AA dosages of 3–15%. The liquid-to-solid ratio (L/S) was calculated using the combined mass of the SH solution (10 M, 40% NaOH by mass) and the SS solution, relative to the mass of the solid binder (CS + RHA = 100 gm basis). For the activator masses employed in Mix-1 through Mix-5 (SH: SS = 1:2, 2:4, 3:6, 4:8, 5:10 gm per 100 gm binder), the resulting L/S values were 0.045, 0.090, 0.135, 0.180, and 0.225, respectively. These L/S values were selected to balance workability and reaction efficiency. Previous studies commonly adopted activator/binder ratios of 0.15–0.25 and SH: SS ratios of 1:2 or 1:2.5 [27, 44]. In the present study, emphasis was placed on optimizing activator dosage and reducing alkali consumption to minimize cost and environmental impact. The adopted proportions of CS, RHA, and AA were followed as per earlier findings [3, 29, 42, 44–47].

Table 4 Mix proportion of composite mix (CS, RHA, and AA)

Combinations	Mixed proportions	Mix Designation	Alkali activator		Tests performed	
			SH (gm)	SS (gm)	Sp. Gravity, Compaction, UCS, and STS tests	FE-SEM/EDX and XRD analysis
Mix-1	10%RHA+90%CS+3%AA	M1A1	1	2	✓	X
	20%RHA+80%CS+3%AA	M1A2	1	2	✓	X
	30%RHA+70%CS+3%AA	M1A3	1	2	✓	X
	40%RHA+60%CS+3%AA	M1A4	1	2	✓	X
	50%RHA+50%CS+3%AA	M1A5	1	2	✓	X
Mix-2	10%RHA+90%CS+6%AA	M2A1	2	4	✓	X
	20%RHA+80%CS+6%AA	M2A2	2	4	✓	X
	30%RHA+70%CS+6%AA	M2A3	2	4	✓	X
	40%RHA+60%CS+6%AA	M2A4	2	4	✓	X
	50%RHA+50%CS+6%AA	M2A5	2	4	✓	X
Mix-3	10%RHA+90%CS+9%AA	M3A1	3	6	✓	X
	20%RHA+80%CS+9%AA	M3A2	3	6	✓	X
	30%RHA+70%CS+9%AA	M3A3	3	6	✓	X
	40%RHA+60%CS+9%AA	M3A4	3	6	✓	X
	50%RHA+50%CS+9%AA	M3A5	3	6	✓	X
Mix-4	10%RHA+90%CS+12%AA	M4A1	4	8	✓	X
	20%RHA+80%CS+12%AA	M4A2	4	8	✓	]
	30%RHA+70%CS+12%AA	M4A3	4	8	✓	X
	40%RHA+60%CS+12%AA	M4A4	4	8	✓	X
	50%RHA+50%CS+12%AA	M4A5	4	8	✓	X
Mix-5	10%RHA+90%CS+15%AA	M5A1	5	10	✓	X
	20%RHA+80%CS+15%AA	M5A2	5	10	✓	
	30%RHA+70%CS+15%AA	M5A3	5	10	✓	X
	40%RHA+60%CS+15%AA	M5A4	5	10	✓	X
	50%RHA+50%CS+15%AA	M5A5	5	10	✓	X

Preparation of samples and testing procedure

CS and RHA were first oven-dried at 105 ± 2 °C for 24 h to remove residual moisture and then stored in airtight containers to prevent reabsorption. The total weight of the dry mix (W) was calculated as $W = W_{(CS)} + W_{(RHA)}$, where $W_{(CS)}$ and $W_{(RHA)}$ denote the dry weights of CS, and RHA, respectively. Further, the additional required percentage of AA was added in the base material. For the representative mix containing 10% RHA, 90% CS, and 15% AA, for a total base material of 400 gm was comprising 40 gm of RHA, 360 gm of CS. The 60 gm of AA (SH and SS) consisted of 20 gm SH (10 M solution) and 40 gm SS (containing 3.20 gm Na_2O , 10.60 gm SiO_2 , and 26.2 gm H_2O). The activator solution was added gradually to the dry CS-RHA mixture and blended using a mechanical stirrer for approximately 5 min to ensure uniformity. The homogeneous mix was then placed into cylindrical moulds (for UCS and STS) and compacted to maintain the density. The UCS samples were conducted on cylindrical samples with a diameter of 38 mm and a height of 76 mm (maintaining a height-to-diameter ratio of 2). After 24 h, the samples were demoulded and cured under ambient conditions (temperature of 25 ± 2 °C and relative humidity of 50 ± 2 °C) until the designated curing ages of 3, 7, and 28 days. All tests were performed in accordance with the relevant IS standards, IRC, and ASTM, as detailed in their respective sections.

Testing methods

CS, and RHA were mixed with AA at various percentages to study the mechanical behavior of the composite geopolymer material. Tests performed included the specific gravity test, standard Proctor compaction test, UCS test, split tensile strength test and life cycle assessment. Additionally, the microstructural and morphological behavior of the composite materials (tested samples) was analyzed using FESEM/EDX and XRD. The standards and sample sizes are discussed below. The specific gravity test was conducted using pycnometer (ASTM D854-14 [48]) and the compaction test was performed in accordance with ASTM D1557-09 for both raw and composite materials [49]. For each material and composite combination, three tests were performed, and the average of these values was considered the final specific gravity. The maximum dry density and optimum moisture content were calculated for each mix (Mix-1 to Mix-5). Various tests and methodologies have been established to evaluate soil tensile strength and overall stability [50]. In the present work, both unconfined compressive strength (UCS) test and split tensile strength test (STS) were conducted to evaluate the strength of the composite materials. The tests

were carried out following ASTM D2166-98 [51] for composite mixtures Mix-1 to Mix-5. The samples for the STS test were prepared in the same way as those for the UCS test. The specimens were placed horizontally between two mild steel strips positioned on the top and bottom bearing surfaces of a strain-controlled compression machine, operating at a rate of 1.25 mm/min. The mild steel (MS) strips measured 5 mm in thickness, 5 mm in width, and 50 mm in length. These strips were provided to ensure uniform pressure distribution. These tests were conducted in accordance with ASTM C496-11 (ASTM 2011b) [52].

The tensile strength was calculated by:

$$T = \frac{2P}{\pi \times L \times D}$$

Where P_{max} = peak load, L =Sample length, D =Sample diameter, T =Tensile strength of specimen.

Life cycle assessment

The Life Cycle Assessment (LCA) was conducted in compliance with ISO 14,040 [53] and ISO 14,044 [54] standards to evaluate and compare the environmental performance of composite mixes incorporating copper slag (CS) and rice husk ash (RHA) with alkali activator and cement binders. Material quantities were estimated for a subgrade pavement section measuring 2 km in length, 7 m in width, and 0.5 m in thickness. The mixes analyzed included 20%RHA + 80%CS + 3%AA and 20%RHA + 75%CS + 5%C, where the cement-based composite properties were adopted from the authors' earlier work [4]. LCA modeling was performed using SimaPro v9.0[®] coupled with the Ecoinvent v3.8 database, and the environmental impacts were quantified using the USEPA TRACI 2.1 impact assessment method [55]. A cradle-to-gate system boundary, consistent with the approach described by Mishra et al., was applied [56]. The evaluation covered four key environmental domains (I) air, (II) water, (III) human health, and (IV) land encompassing ten midpoint impact indicators: ozone depletion, global warming, smog formation, acidification, eutrophication, carcinogenic and non-carcinogenic effects, respiratory impacts, ecotoxicity, and fossil fuel depletion.

Microstructure analysis

The microstructural tests were conducted to identify the existence of strength-improving compounds in the composite material using scanning electron microscope (SEM) and X-ray diffraction (XRD) analysis. Only the optimum UCS tested samples (after 28 days of curing) were examined. The details of the tests are as follows:

Field emission-scanning electron microscope and EDX tests

The field emission scanning electron microscope (FE-SEM) is a high-resolution imaging tool capable of capturing highly magnified images of materials, providing detailed insights into their microstructural morphology, including particle size, composition, and surface characteristics. The SEM analysis was performed using a Zeiss Supra 55 microscope, operating at a high resolution of 3 nm. Micrographs were captured at magnifications 5k, 10k and 50kx. Subsequently, EDX mapping was carried out to determine the elemental composition and spatial distribution of elements within the hydrated or tested samples. In this study, the morphology of the geo-polymerized copper slag (CS) and rice husk ash (RHA) mixture was examined using FE-SEM and energy-dispersive X-ray spectroscopy (EDX). To improve the electrical conductivity of the composite samples (tested under UCS), small specimens were extracted from the cylindrical molds of the optimum UCS samples. These samples, consisting of CS and RHA mixtures activated with AA, were affixed to double-sided carbon tape, sputter-coated, and mounted on metallic stubs with a thin layer of gold to prepare them for imaging.

X-ray diffraction (XRD) analysis

Following the determination of UCS values, the XRD tests were conducted on samples of the optimal blend using the PERT PRO, a multi-purpose diffractometer manufactured by Malvern Analytical. The analysis was performed under fixed parameters, including an observation angle range of 10° – 80° , a step size of 0.008° , a scan rate of $0.5^{\circ}/\text{min}$, and a scan step duration of 10.16 s. To prepare the samples, the UCS-tested specimens were ground into a fine powder using a grinding machine for 5 min, then sieved through a $45\text{ }\mu\text{m}$ sieve. Twenty grams of the dry, sieved material were transferred in an airtight sample poly bag for analysis. The XRD data were processed using High Score software, with the phases of the substances identified through the International Centre for Diffraction Data (ICDD). Table 4 presents the details of the optimal mixtures used for XRD analysis.

Results and discussion

The investigation of the test results and the discussion are presented in the following sections:

Specific gravity and compaction properties of composite mixes

The specific gravity (G_s) values of different composite mix proportions are presented in Figs. 2a to e. CS exhibited a specific gravity of 3.50, which falls within the reported range of 3.37–3.97 [57], while RHA showed a lower value of 2.06. The higher specific gravity of CS is attributed to its high iron oxide content (40.77%) compared to that of RHA (1.07%). The combined specific gravity of the AA (SH and SS) was calculated as 1.44. It was observed that the incorporation of RHA and AA into CS reduced the specific gravity of all composite mixes (Mix-1 to Mix-5). Among these, the maximum specific gravity of 3.47 was obtained for the composite mix containing 10%RHA, 90%CS, and 15%AA (M5A1), while the minimum value of 2.72 was recorded for the mix with 50%RHA, 50%CS, and 3% AA (M1A5). The decrease in specific gravity is attributed to the reduction in iron content caused by the addition of RHA and AA, as well as the incorporation of finer, lightweight particles into the coarser CS matrix [27, 44].

Figures 2a to e show the compacted maximum dry unit weight versus water content curves of the composite mixes (Mix-1 to Mix-5). The results indicate that with an increase in RHA content from 10% to 50% and a constant percentage of AA, the composite material exhibited a decrease in maximum dry unit weight and a corresponding increase in optimum moisture content (OMC). The observations for combinations of RHA, AA, pond ash (PA), and CS treated with lime, cement, and alkali activators are consistent with previous research findings [3, 27].

Among all composite mixes, the highest and lowest maximum dry unit weights (17.35 kN/m^3 and 5.45 kN/m^3) and OMC values (15.35% and 22.52%) were obtained for the mixes containing 10%RHA, 90%CS, and 3%AA, and 50%RHA, 50%CS, and 15%AA, respectively. The decrease in MDD and the corresponding increase in OMC with increasing RHA and AA content are primarily attributed to the lightweight (lower specific gravity) and porous nature of RHA and its high water-retention capacity between particles. As the RHA proportion increases, the composite becomes more difficult to compact due to reduced particle density and greater moisture demand. Conversely, variations in AA content influence compaction behavior due to its liquid characteristics, which delay early particle binding. The AA reacts gradually with the raw materials rather than forming an immediate bond, while partially substituting for water during compaction. These observations confirm that the developed green composite incorporating CS, RHA, and AA as a cement substitute demonstrates favorable compaction behavior and can be effectively utilized as a sustainable material for flexible pavement subgrade applications.

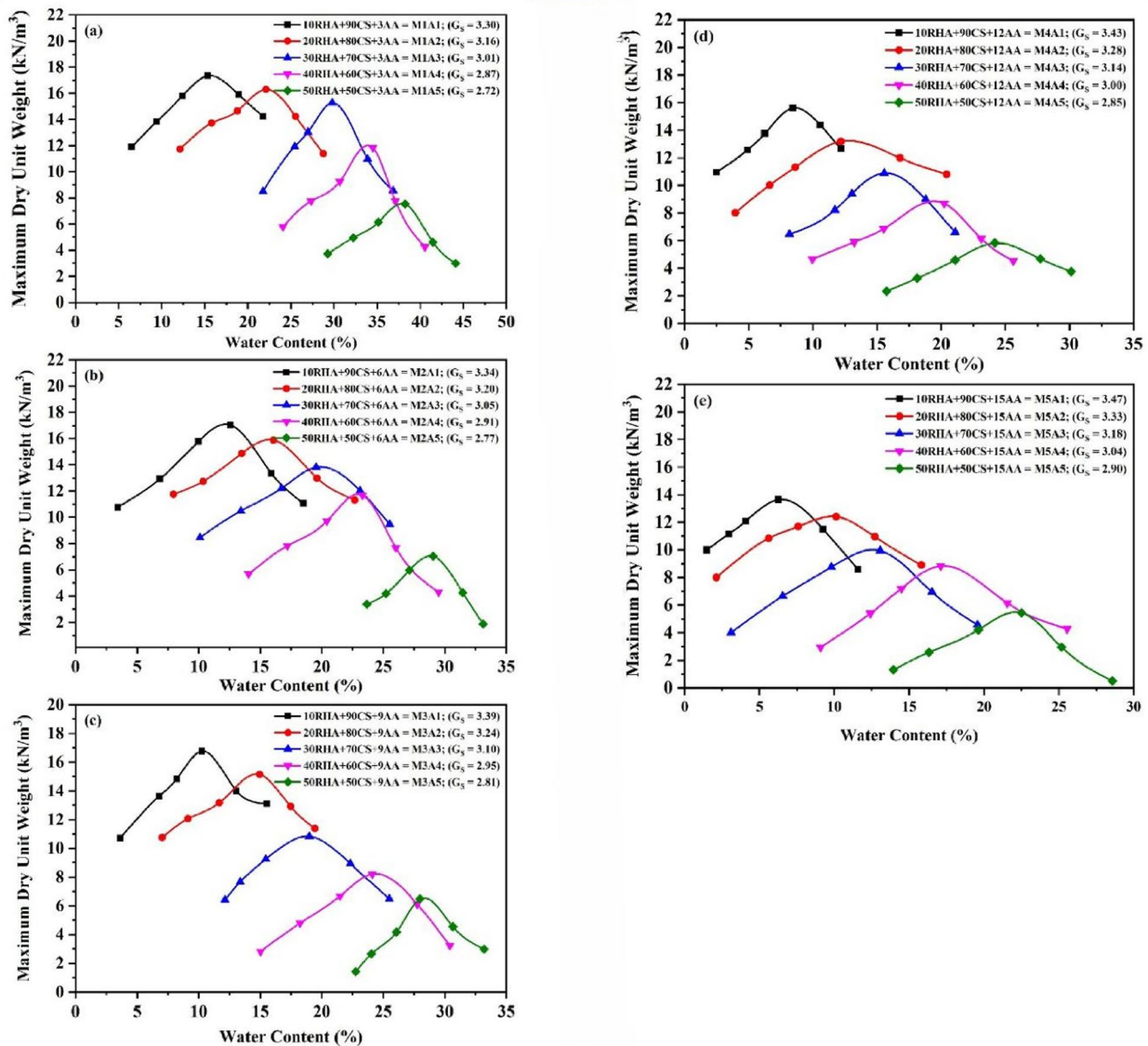


Fig. 2 Relationship between MDD and OMC for different mixes: (a) Mix-1, (b) Mix-2, and (c) Mix-3, (d) Mix-4 and (e) Mix-5

Unconfined compressive strength of composite mixes

Stress-strain behavior

The stress-strain behavior of composite mixes (M1-M5) after 28 days of curing is presented in Figs. 3a to e. All mixes exhibit an initial linear relationship between stress and strain up to about 1.18% strain, reflecting elastic deformation. At low strain levels (up to 0.4%), the stress-strain curves of all composites show minimal variation, and indicating comparable stiffness in the elastic region. Beyond this range, stress increases until reaching a distinct peak, representing the material's ultimate load-bearing capacity, followed by a sharp drop in stress signifying brittle post-peak failure. Such

behavior is typical of geopolymer composites with strong but rigid bonding, where failure occurs abruptly with negligible plastic deformation.

The stress-strain responses for varying RHA contents (10–50%) and fixed AA levels (3–15%) show a consistent trend across all mixes. For each activator level (3, 6, 9, 12, and 15%), the composite with 20% RHA and 80% CS achieved the peak strength, while higher RHA contents resulted in reduced performance. The maximum stress values increased progressively with higher percentage of AA from 5527.67 kPa (3%AA) to 10,234.47 kPa (15%AA) indicating enhanced geopolymerization and gel formation. Conversely, the lowest strengths were consistently observed in mixes with 50% RHA and 50% CS, confirming dilution of reactive constituents at higher RHA levels. An excess Si/

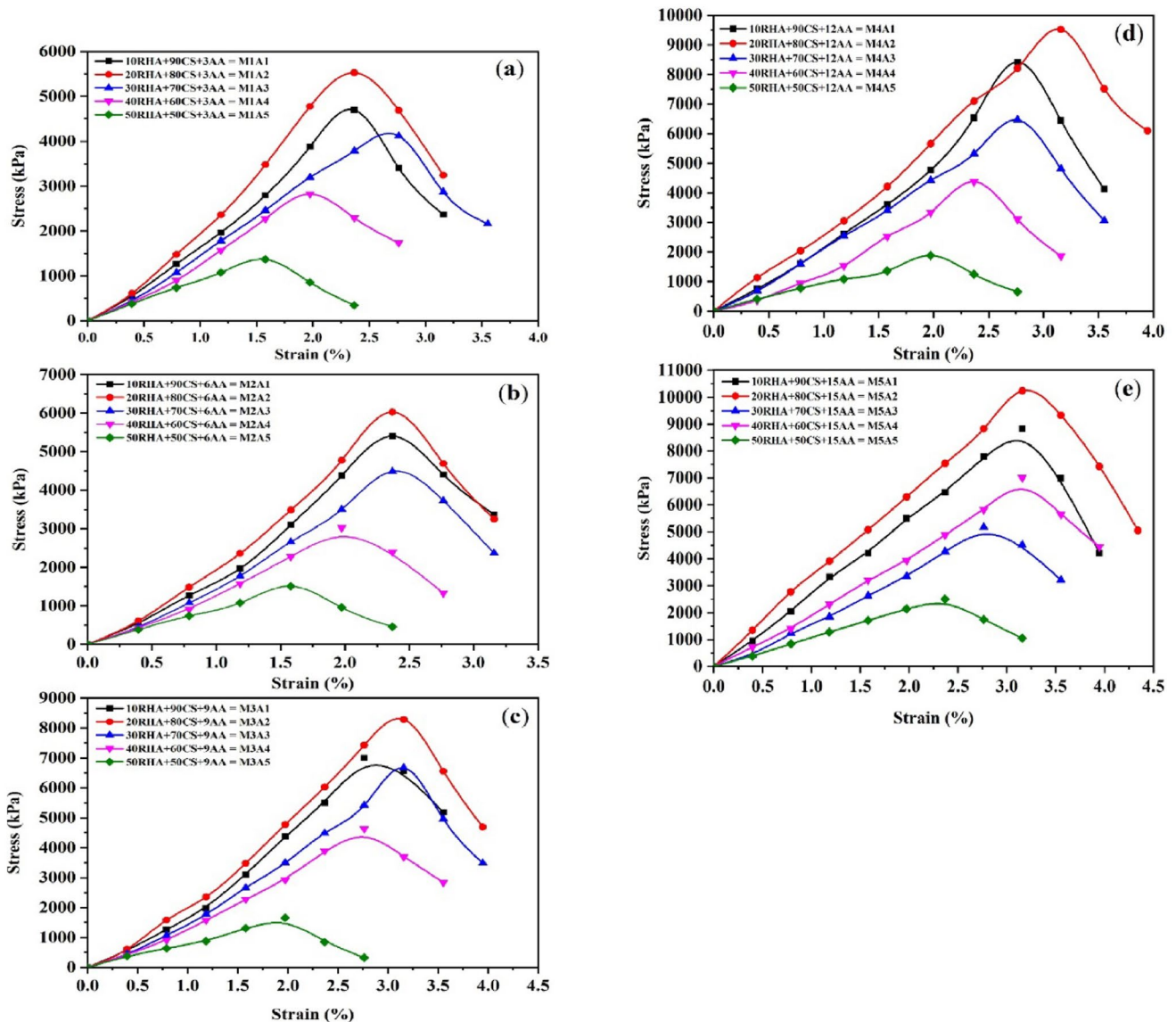


Fig. 3 Stress-strain behavior of composite: (a) Mix-1, (b) Mix-2, (c) Mix-3, (d) Mix-4 and (e) Mix-5 at 28 days

Al ratio limits the development of balanced aluminosilicate gel phases, further contributing to the brittle response of the composites.

Effect of curing period

The variations of UCS values of the composite mixes at 3, 7, and 28 days of curing are shown in Figs. 4a to e. Across all activator percentage levels, UCS consistently increased with curing age due to progressive geopolymerization and matrix densification. For each AA concentration, the optimum performance was observed at 20%RHA and 80%CS, whereas higher RHA contents resulted in a decrease in UCS because excess silica disrupted the geopolymer network.

At 3% AA (Fig. 4a), UCS increased with RHA content up to 20%, after which a noticeable decline occurred. The optimum mix, M1A2, exhibited a significant strength gain of 106.60% from 3 to 28 days, reaching 5527 kPa. A similar trend was observed at 6%AA (Fig. 4b), where M2A2 attained the highest UCS of 6027.67 kPa, improving by 87.45% with curing. Beyond the 20%RHA threshold, UCS values decreased due to excessive siliceous phases limiting gel formation.

At 9% AA (Fig. 4c), UCS again peaked at 20%RHA, with M3A2 reaching 8287.32 kPa a 128.88% increase over the curing period. Strength development remained strongly influenced by extended curing, which enhanced aluminosilicate gel formation. At 12%AA (Fig. 4d), M4A2 achieved 9524.80 kPa after 28 days, showing a 117.39% improvement

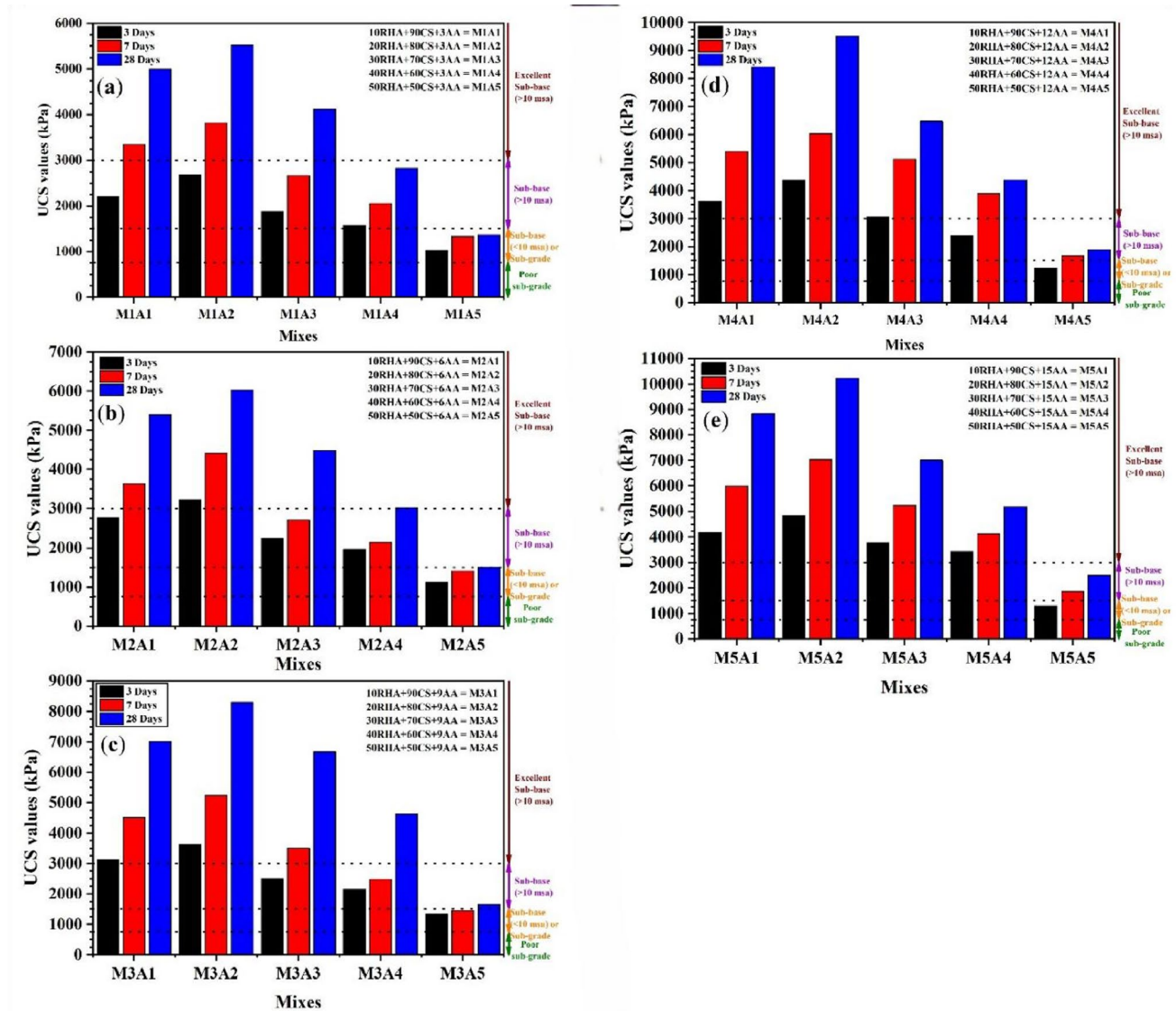


Fig. 4 Variation of UCS values with various curing periods: (a) Mix-1, (b) Mix-2, (c) Mix-3, (d) Mix-4, and (e) Mix-5

from early-age curing. At 15% AA (Fig. 4e), M5A2 recorded the highest UCS among all mixes at 10234.47 kPa, with a 113.13% gain over the curing period.

The consistent peak at 20% RHA across all activator percentage levels highlights the balanced availability of reactive silica and alumina necessary for effective geopolymerization. Increasing AA concentration from 3% to 15% significantly enhanced UCS by improving precursor dissolution and accelerating N-A-S-H and C-A-S-H gel formation. However, RHA contents beyond 20% introduced excess silica, which disrupted the Si/Al ratio and produced a less cohesive matrix. Chemically, the NaOH and Na_2SiO_3 activator dissolves amorphous silica and alumina from RHA and CS, initiating the formation of a three-dimensional aluminosilicate network. Extended curing provides adequate time for these gels to polymerize and densify the

matrix, resulting in higher UCS. Thus, optimum strength is achieved through the combined effects of adequate AA percentage, controlled RHA content, and longer curing duration. The effect of RHA, AA, and curing period are strongly supported by the results, as well as the percentages of materials used [16, 40, 47, 58–61].

According to IRC: 37-2018 [62] and IRC: SP-72-2015 [63] guidelines, the UCS of pavement layers serves as a key indicator of their suitability and long-term performance under varying traffic conditions. For low to medium traffic volumes (< 10 million standard axles, MSA), the UCS requirement ranges between 0.75 and 1.5 MPa for subgrade and sub-base layers after a 7-day curing period. For higher traffic intensities (> 10 MSA), a higher UCS of 1.5 to 3.0 MPa is mandated for sub-base layers to ensure adequate load distribution and deformation resistance. In the present study,

the performance of various composite mixes was evaluated with reference to these strength criteria. For composite Mix-1, Mix-2, and Mix-3 containing 3%, 6%, and 9% alkali activator (AA), respectively, all mixes achieved the minimum required UCS range for subgrade and sub-base applications under traffic flow < 10 MSA after 7 days of curing (Figs. 4a to c). Among these, M1A1 to M1A4 (for Mix-1); M2A1 to M2A4 (for Mix-2); and M3A1 to M3A4 (for Mix-3) exhibited superior strength values, meeting the higher UCS range prescribed for sub-base layers under heavy traffic conditions (> 10 MSA). Similarly, all mixes from Mix-4 and Mix-5 satisfied IRC strength requirements for both traffic categories (< 10 MSA and > 10 MSA). Among all mixes, M1A2 has the most cost-effective (minimum percentage AA used) while meeting all performance thresholds.

Figures 5a and b represent the graphs plotted between peak strain energy and peak strength, indicating the balance between material strength and brittleness. The energy absorption capacity (E_u) and elastic stiffness modulus (E_{50}) were measured for all mixes (Mix-1 to Mix-5) at different curing periods to assess toughness and stiffness behaviour. The results show that peak strain energy follows a trend similar to peak strength, increasing with curing age, mix proportion, and alkali activator content. A strong correlation

was observed between energy absorption capacity (E_u) and peak strength, expressed as $E_u = 2.175q_u$ with an R^2 value of 0.92. Likewise, the correlation between elastic stiffness modulus (E_{50}) and peak strength was $E_{50} = 0.235q_u$ ($R^2 = 0.95$). These relationships are consistent with the findings reported by Dixit et al. (2024) [64], further validating the mechanical performance of the CS-RHA-AA composite mixes.

Split tensile strength test of composite mixes

Stress-strain behavior

Figures 6a to e present the split tensile strength (STS) strain behavior of the composite mixes (Mix-1 to Mix-5) after 28 days of curing. All mixes initially show a linear increase in tensile stress with strain up to about 0.75%, indicating elastic deformation during early loading. Beyond this region, each mix achieves a distinct peak tensile strength followed by a post-peak softening phase, which is characteristic of geopolymer composites. Across all AA percentage levels, the highest tensile performance consistently occurred at the composition containing 20%RHA and 80%CS. The peak STS values recorded were 53.50 kPa for M1A2 (3%AA), 68.56 kPa for M2A2 (6%AA), 74.64 kPa for M3A2 (9%AA), 80.56 kPa for M4A2 (12%AA), and 98.24 kPa for M5A2 (15%AA), with strain capacities ranging from 1.58% to 2.37%. Similar to the UCS value, mixes with 50%RHA consistently showed the lowest STS values 14.81 kPa to 25.81 kPa measured at 1.18% strain, confirming that excessive RHA weakens bonding efficiency and reduces mechanical performance.

The AA promotes dissolution of silica (SiO_2) and alumina (Al_2O_3) from RHA and CS, enabling the formation of geopolymeric gels such as N-A-S-H and C-A-S-H, which serve as the primary binding phases [65]. At 20%RHA, the silica-alumina ratio is optimally balanced with the metallic (iron oxide) in CS, producing a dense and cohesive gel network capable of redistributing stresses effectively, leading to improved tensile strength and strain tolerance. When RHA content exceeds 20%, excess silica disrupts the Si/Al ratio, leaving unreacted particles that weaken the gel matrix and increase brittleness [66, 67]. Increasing AA content (9 to 15%) further enhances tensile performance by improving precursor dissolution and promoting the development of a more interconnected gel structure [68]. The combination of 20% RHA, 80% CS, and appropriate AA percentage yields a composite with superior tensile strength, improved ductility, and greater resistance to strain-induced failure [31].

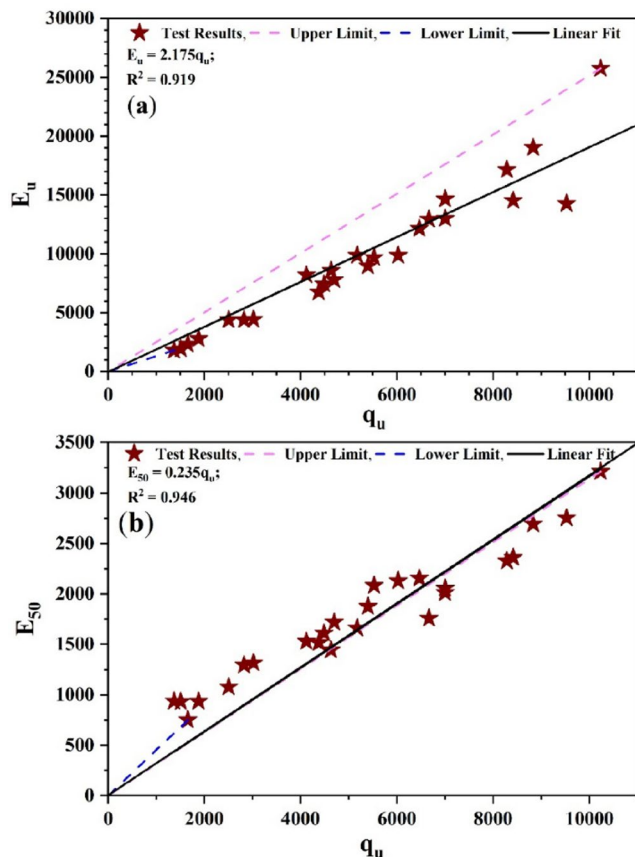
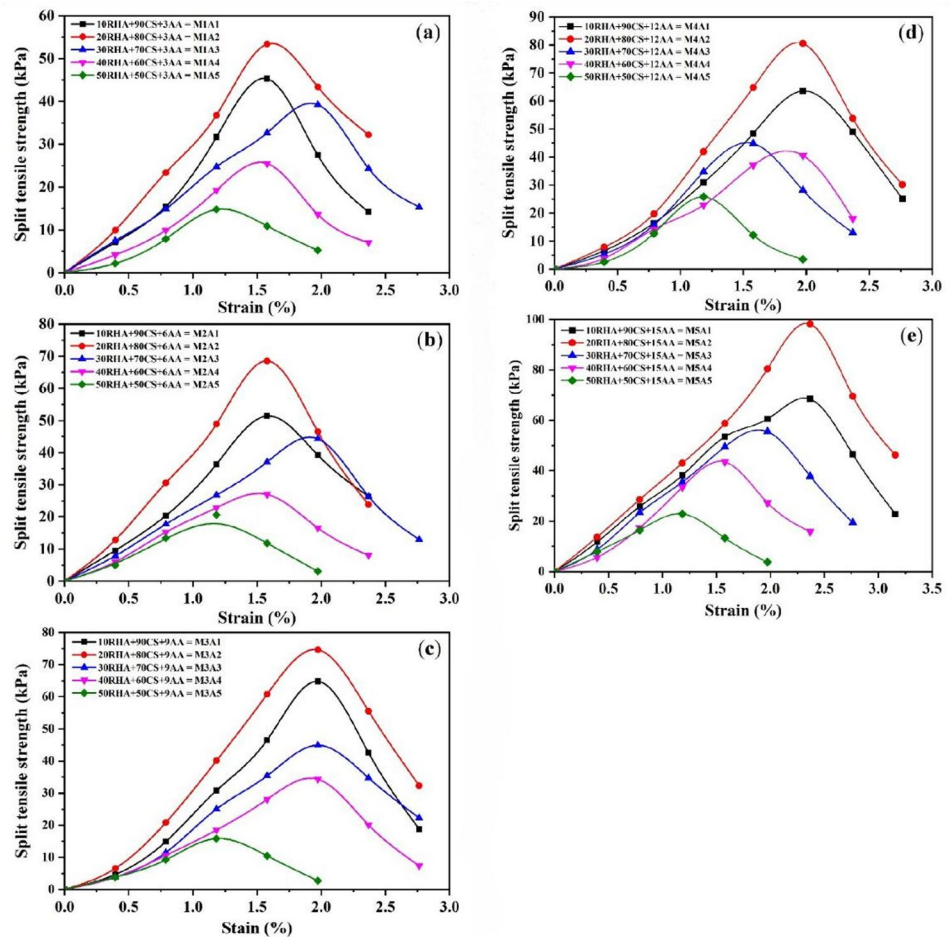


Fig. 5 Relation between (a) E_u and q_u ; (b) E_{50} and q_u

Fig. 6 STS-strain response of (a) Mix-1, (b) Mix-2, (c) Mix-3, (d) Mix-4, and (e) Mix-5 at 28 days of curing



Effect of curing periods

The STS of composite mixes containing varying proportions of RHA and CS was evaluated across five series (Mix-1 to Mix-5) with alkali activator (AA) levels of 3%, 6%, 9%, 12%, and 15%, as illustrated in Fig. 7a to e. Tensile strength increased consistently with curing time and AA percentage up to 28 days for all mixes. In each series, the optimal performance was obtained at a composition of 20%RHA and 80%CS, with mixes such as M1A2, M2A2, M3A2, M4A2, and M5A2 showing the highest STS and UCS values. These mixes exhibited substantial improvements in tensile strength, ranging from 64% to 77% over the curing period. Strength increased as RHA content up to 20%; however, further increases (>20%) led to a decline in performance due to increased porosity, reduced cohesion, and weakened bonding efficiency. Early-age results also indicated that mixes with $\leq 20\%$ RHA achieved better early reaction rates and matrix development. Additionally, higher AA percentage improved overall performance, with M5A2 (15%AA) achieving the maximum STS of 98.24 kPa at 28 days. These findings confirm that a mix containing 20%RHA and suitable AA percentage levels provides optimal early and

long-term tensile strength, while excessive RHA content negatively affects structural integrity. The improvement in STS with curing time and optimal RHA-CS-AA composition results from the geopolymerization reactions and the pozzolanic activity of RHA.

Environmental impact assessment

The environmental impact assessment per MPa of sub-grade strength indicates that the alkali-activated mix (20%RHA+80%CS+3%AA) performs better overall than the cement-base mix (20% RHA+75% CS+5%C), as shown in Fig. 8. For ozone depletion, alkali activator (AA) emits 0.00197 kg CFC-11 eq per MPa, approximately half of the 0.00420 kg CFC-11 eq per MPa from the cement mix, indicating lower industrial emissions. The global warming potential of AA is 27,725.8 kg CO₂ eq per MPa, about 4.3 times lower than 120,674.6 kg CO₂ eq per MPa from cement, due to reduced clinker use. Smog formation is 1,625.5 kg O₃ eq per MPa for AA versus 5,767.6 kg O₃ eq per MPa for cement, showing lower photochemical pollution. Acidification is also reduced for AA (188.0 kg SO₂ eq per MPa) compared to cement (280.5 kg SO₂ eq per MPa),

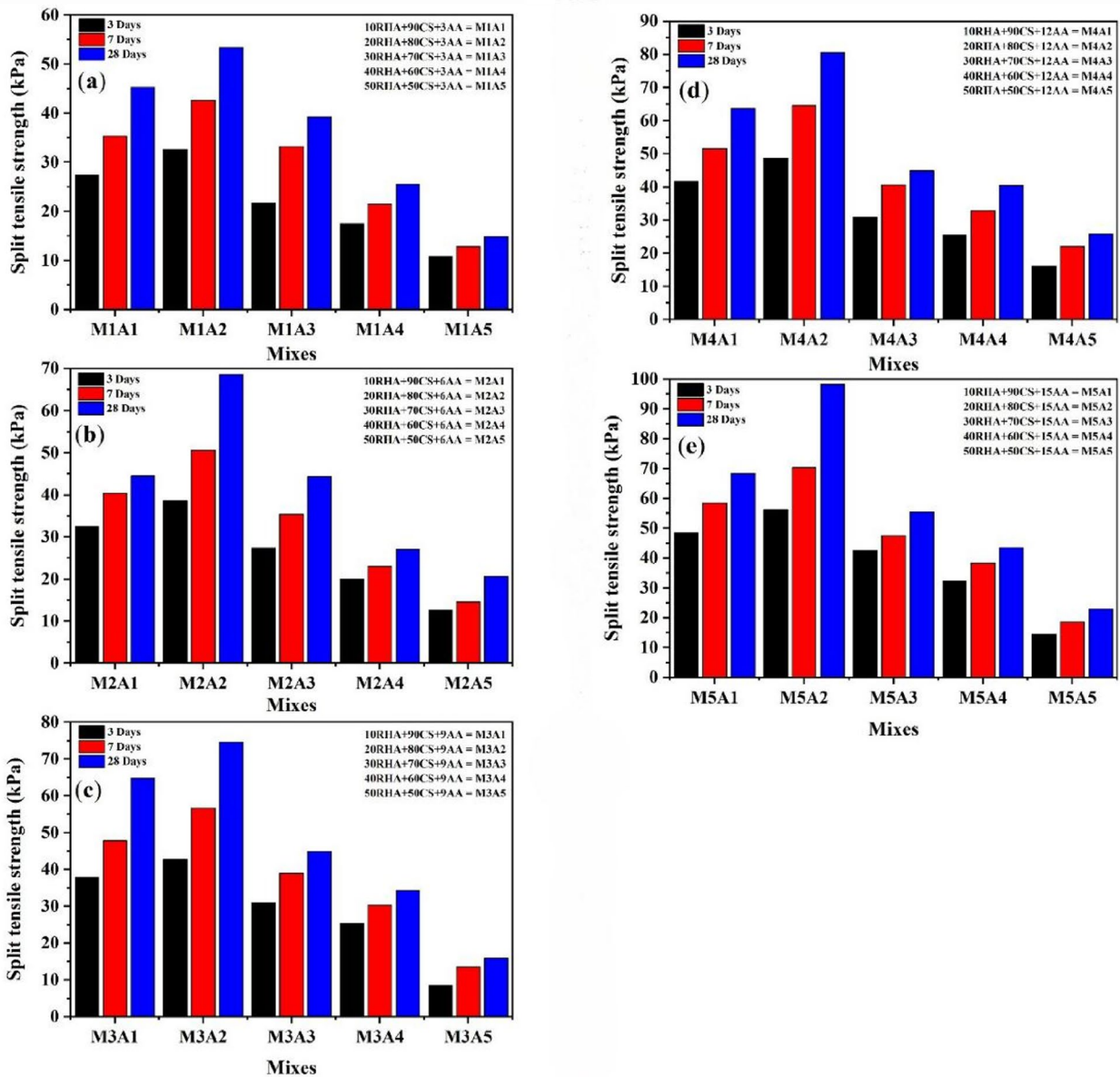


Fig. 7 Effect of curing period on the STS of (a) Mix-1, (b) Mix-2, (c) Mix-3, (d) Mix-4, and (e) Mix-5

and eutrophication is lower (10.57 kg N eq per MPa for AA vs. 26.18 kg N eq per MPa for cement), reflecting decreased nutrient release. Carcinogenic impacts are nearly identical (0.000559 CTUh per MPa for AA and 0.000555 CTUh per MPa for cement), while non-carcinogenic toxicity is lower for AA (0.00398 CTUh per MPa) than cement (0.00623 CTUh per MPa). Respiratory effects are 19.34 kg PM_{2.5} eq per MPa for AA compared to 32.60 kg PM_{2.5} eq per MPa for cement, indicating reduced risks to human health. However, ecotoxicity is higher for AA (23,776.2 CTUe per MPa) than cement (17,867.2 CTUe per MPa), likely due to heavy metals in copper slag and other industrial byproducts. Finally, fossil fuel depletion is 26,825.3 MJ per MPa for AA, considerably lower than 46,864.9 MJ per MPa for

cement, reflecting lower energy requirements. Overall, on a per-MPa basis, the AA mix significantly reduces environmental burdens such as greenhouse gas emissions, air pollution, acidification, eutrophication, respiratory impacts, and energy consumption, while careful handling of industrial byproducts is required to manage ecotoxicity.

Microstructure behaviour composite mix

FE-SEM and EDX mapping

The FE-SEM images provide detailed insight into the microstructure of the CS-RHA-AA geopolymer composite,

Fig. 8 Environmental midpoint impact assessment of alkali activator based composite (20%RHA+80%CS+3%AA) compared to cement based composite (75%CS+20%RHA+5%C) using life cycle assessment

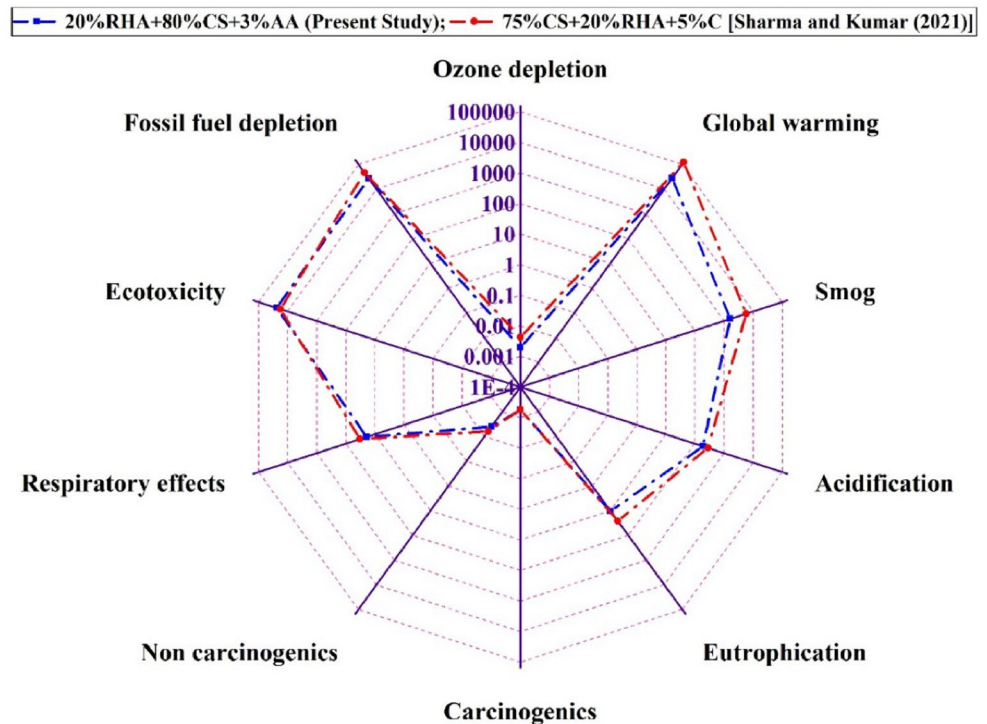
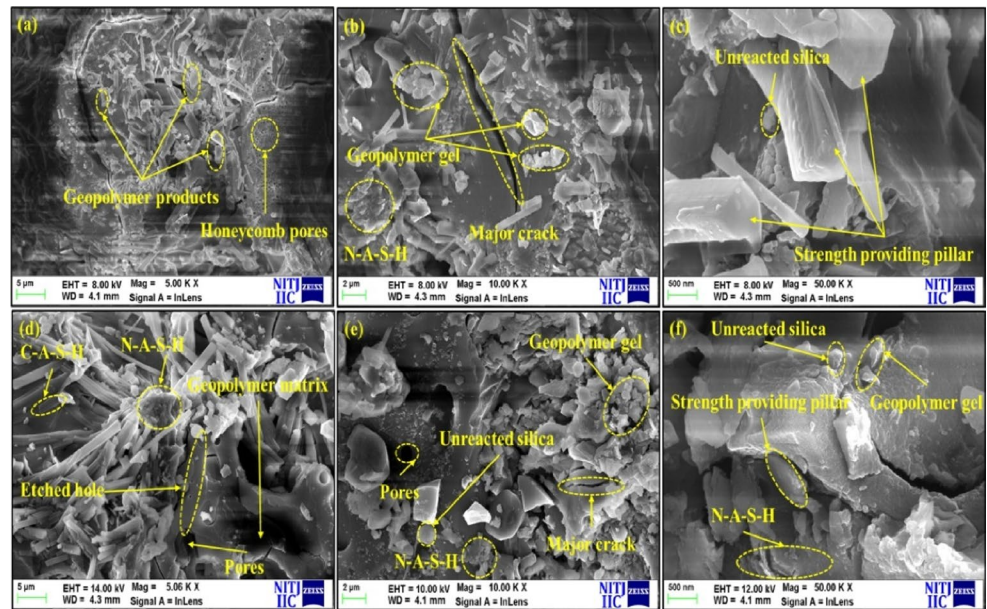


Fig. 9 FE-SEM images of optimum UCS tested samples at 28 days of curing (a-f)



highlighting features that directly influence its mechanical and chemical performance, as shown in Figs. 9a to f.

Figure 9a reveals geopolymer reaction products and honeycomb-like pores formed through localized dissolution of precursor materials primarily rice husk ash (RHA) during alkali activation. This dissolution releases reactive silicate and aluminate species that subsequently polymerize to form the aluminosilicate network. In Fig. 9b, well-developed geopolymeric gel phases, predominantly sodium aluminosilicate hydrate (N-A-S-H), are observed. These gels form

the primary binding matrix, contributing to early strength gain and structural integrity. Although minor microcracks are present, the overall matrix appears dense and interconnected, indicating effective polymerization that enhances load transfer and reduces permeability. Fig. 9c shows unreacted silica particles and semi-crystalline, strength-providing pillars composed of polymerized silicate and aluminate chains, which act as reinforcement phases crucial for maintaining structural continuity and chemical stability. Fig. 9d highlights the coexistence of C-A-S-H gel, N-A-S-H gel,

and crystalline secondary phases, accompanied by etched voids and limited porosity. Their interconnected morphology contributes to improved rigidity, fracture resistance, and stress-transfer efficiency within the matrix. Fig. 9e and f further illustrate matrix consolidation, with Fig. 9e showing a compact N-A-S-H-dominated gel network that restricts crack propagation and enhances stiffness, while minor crystalline inclusions provide additional reinforcement. Fig. 9f emphasizes internal porosity and load-bearing pillars, where pores likely arise from incomplete particle packing, moisture evaporation, or trapped air during mixing. Isolated micro pores have minimal influence on performance, whereas larger or interconnected pores reduce compressive strength and increase permeability; however, controlled porosity can be advantageous for lightweight or permeable applications.

The FE-SEM observations confirm that the composite forms a dense, cohesive, and chemically stable microstructure governed by N-A-S-H and C-A-S-H gel development. CS, rich in Fe_2O_3 , SiO_2 , and Al_2O_3 , enhances packing density and provides a rigid skeleton, while RHA supplies reactive amorphous silica that promotes additional gel formation. The alkali activator (NaOH and Na_2SiO_3) accelerates dissolution and geopolymerization, creating a durable aluminosilicate network. Together, these constituents yield

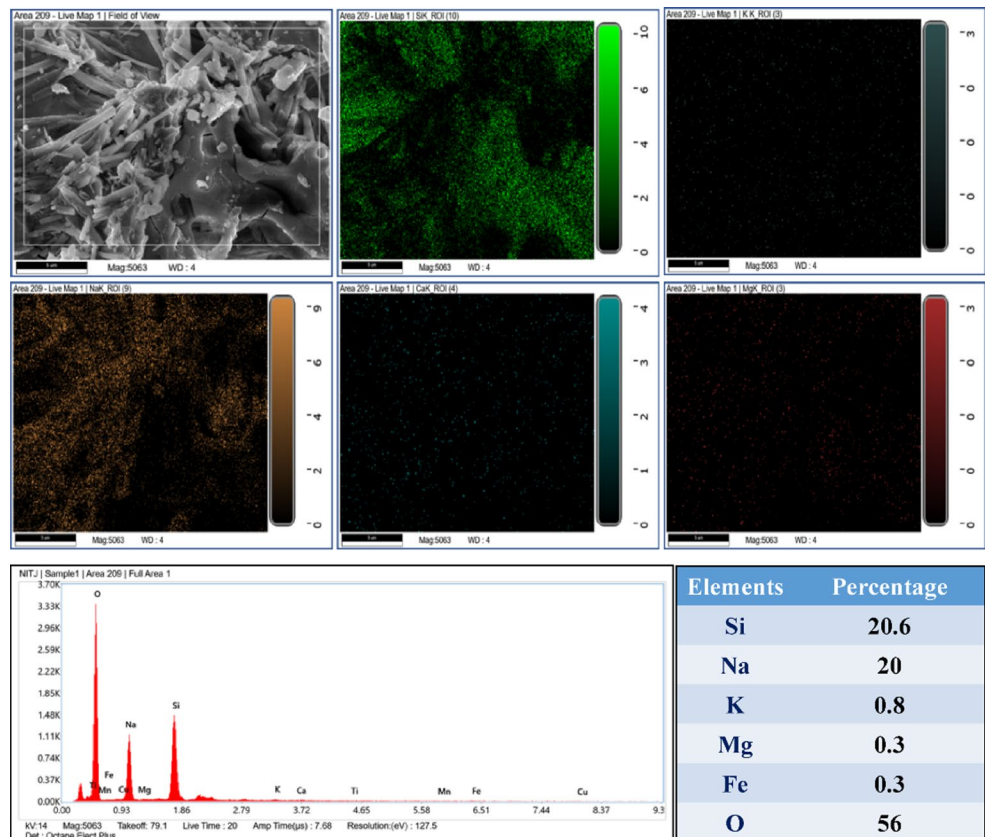
a dense, stable, and sustainable composite suitable for pavement subgrade applications.

Figure 10 presents the Energy-Dispersive X-ray Spectroscopy (EDX) analysis of the CS-RHA-AA geopolymer composite, providing detailed information on its elemental composition and chemical characteristics. The EDX spectrum shows prominent peaks of Si, Na, K, Mg, Fe and O, confirming the presence of key constituents involved in geopolymerization. Strong Si and O peaks indicate the dominance of silicate-based phases, primarily contributed by RHA and CS. The noticeable Na peak reflects the influence of the alkali activator, supporting the formation of sodium aluminosilicate hydrate (N-A-S-H) gels, which serve as the main binding phase in the matrix. The presence of Al further confirms aluminosilicate network formation, while Ca peaks suggest the coexistence of calcium silicate hydrate (C-S-H), enhancing strength and densification. Fe and Mg originate from copper slag, contributing to structural stability. Fig. 10 confirms the chemical composition necessary for a dense, cohesive, and durable geopolymer matrix.

XRD analysis

The hydration behavior and XRD results of the composite mix containing CS, RHA, and AA reveal a complex interplay between crystalline and amorphous phases, which

Fig. 10 EDX spectrum images and hydrated elemental composition



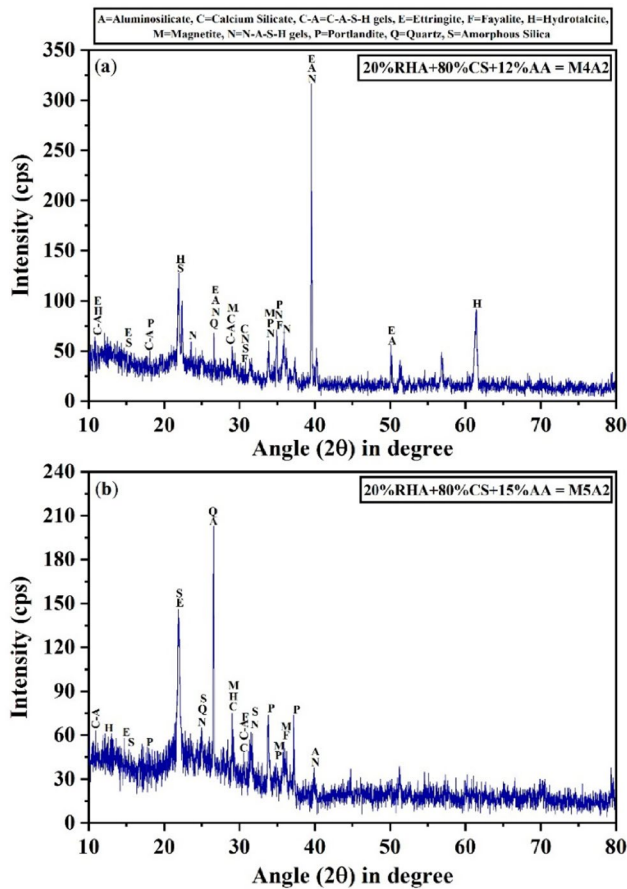


Fig. 11 XRD patterns of optimum mixes (A) M4A2 and (B) M5A2 after 28 days of curing

governs the material's properties, as shown in Figs. 11a and b.

The XRD pattern shows sharp peaks and a broad hump, indicative of a mix of crystalline and amorphous components. Copper Slag, being rich in silica, iron oxides, and alumina, contributes to the observed crystalline peaks. Peaks around 30–35° 2θ likely correspond to crystalline phases like fayalite (Fe_2SiO_4) and magnetite (Fe_3O_4), which are characteristic of CS. These phases remain largely unreacted during the hydration process but act as micro fillers, improving the composite's density and mechanical strength. The peak at 26.52° 2θ, prominent in the XRD results, likely corresponds to quartz, originating from residual crystalline silica in the RHA.

The existence of a broad amorphous hump in the XRD pattern, typically between 15–30° 2θ, reflects the contribution of amorphous silica in the RHA and the formation of geopolymer gels during alkali activation. RHA, being rich in amorphous silica, reacts readily with the AA to form sodium-alumino-silicate-hydrate (N-A-S-H) gels. These gels contribute to early strength development and form the primary binding phase in the composite. The amorphous

nature of the hydration products, such as N-A-S-H and calcium-alumino-silicate-hydrate (C-A-S-H) gels, is consistent with the observed broad hump in the XRD pattern.

AA plays a crucial role in the hydration process, dissolving silica and alumina from the CS and RHA. This dissolution leads to the development of amorphous binding phases like N-A-S-H and C-A-S-H gels, which provide the composite's structural integrity. Additionally, secondary crystalline phases, such as hydrotalcite-like compounds, may form, contributing to long-term strength and reducing porosity. These phases, if present, would appear as less intense but identifiable peaks in the XRD pattern. The hydration behavior and XRD results also demonstrate the dual role of unreacted phases. Crystalline phases like quartz (from RHA) and fayalite or magnetite (from CS) act as micro fillers, improving the material's packing density and mechanical performance. Meanwhile, the reactive amorphous silica from RHA and the glassy phases of CS interact with the AA, forming binding gels that ensure the material's cohesiveness. The XRD results highlight the material's composition, with sharp peaks representing residual crystalline phases and the broad hump indicating amorphous hydration products. The hydration behavior is governed by the interaction of CS, RHA, and the AA, leading to the formation of strength-enhancing compounds such as C-A-S-H gels, ettringite, hydrotalcite, N-A-S-H gels and portlandite. The crystalline phases act as fillers, while the amorphous phases contribute to the binding matrix, resulting in a dense, durable composite material. The combination of these phases provides early strength, long-term durability, and enhanced mechanical performance.

Statistical predictive model

Regression models are presented based on the database developed in the present work for MDD, OMC, UCS and STS based on the percentage of CS, RHA, AA and curing (days) time. The dataset is divided into two parts: 80% of the data is used for training and 20% for testing. The dependent variable is represented by the predicted value, while the observed values acts as independent variables. The SPSS software is used to perform multiple linear regression analysis (MLRA), to validate experimental results [27, 38]. The general MLRA model is expressed as:

$$A = \alpha_o + \alpha_1 B_1 + \alpha_2 B_2 + \dots + \alpha_n B_n + \epsilon \quad (1)$$

Where,

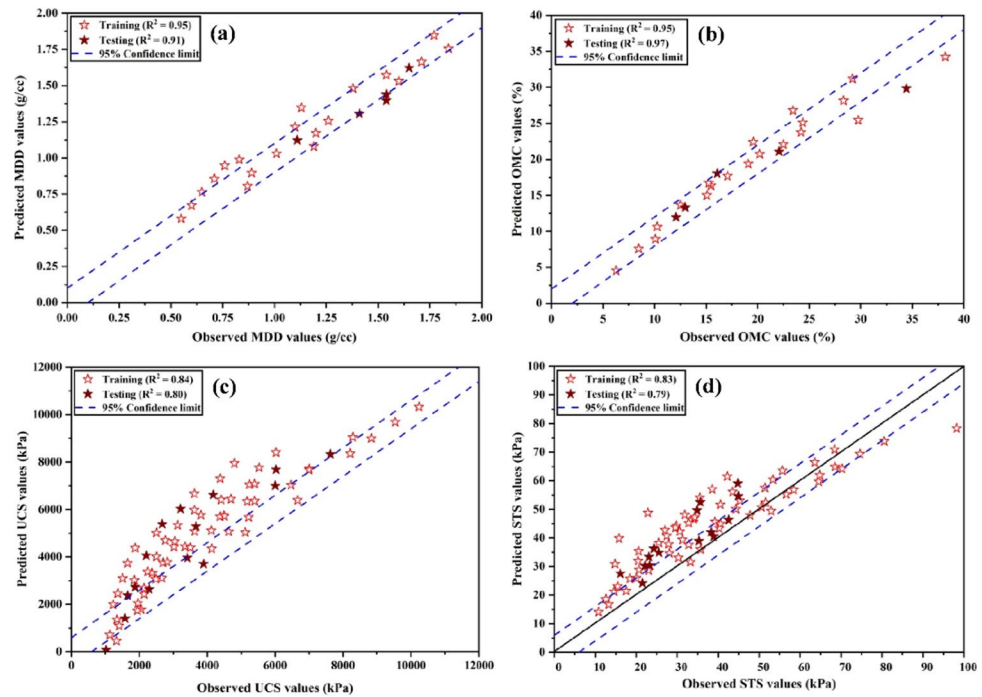
A = Dependent variable.

B_n = Independent variable.

α_o = Intercept constant.

ϵ = Error term.

Fig. 12 Predicted vs. observed values for (a) MDD, (b) OMC, (c) UCS, and (d) STS



Based on the above analysis, the following predictive equations are developed:

$$(MDD)_P = 0.0014 \times RHA + 0.024 \times CS - 0.030 \times AA - 0.23 \quad (2)$$

$$(OMC)_P = 0.0019 \times RHA - 0.436 \times CS - 1.013 \times AA + 58.99 \quad (3)$$

$$(UCS)_P = 116.77 \times RHA - 16 \times CS - 213.286 \times AA - 95.181 \times C - 4188.23 \quad (4)$$

$$(STS)_P = 0.221 \times RH + 0.959 \times CS + 1.489 \times AA + 0.6718 \times C - 51.395 \quad (5)$$

Figure 12a to d illustrate the relationship between predicted and experimentally observed values of MDD, OMC, UCS, and STS. For MDD, R^2 values were 0.95 (training) and 0.91 (testing), while for OMC they were 0.95 and 0.97, indicating excellent agreement. Data points closely align along the 1:1 line within the 95% confidence limits, demonstrating robust generalization, minimal overfitting, and accurate representation of compaction behaviour. The UCS model achieved R^2 values of 0.84 (training) and 0.80 (testing), and the STS model 0.83 (training) and 0.79 (testing), confirming reliable prediction of mechanical responses. The close clustering of predicted versus observed values highlights the models generalization capability. Equations 2–5 may serve as practical predictive tools for sustainable pavement material design of the models for predicting compaction characteristics and

mechanical properties, providing a basis for optimizing mix designs.

Limitations of the present study

The laboratory experiments were conducted under controlled conditions, which may not fully represent field exposure. The durability assessment parameters such as wet-dry, freeze-thaw and abrasion resistance were not studied. The performance evaluation was limited to 28 days of curing, leaving the long-term behaviour, including extended geopolymerization activity and strength development beyond 90 to 365 days, unexplored.

Summary and conclusions

This study investigated the feasibility of using CS and RHA as sustainable materials for pavement construction through alkali activation. The experimental analysis, including compaction tests, UCS tests, split tensile strength tests, and microstructural evaluations (FE-SEM, EDX, and XRD), provided valuable insights into the mechanical and chemical behavior of the developed composite material.

- Increasing RHA content led to a decrease in MDD and an increase in OMC, primarily due to RHA's lower specific gravity and higher water absorption capacity.

Conversely, increasing the AA content reduced OMC across all mixes.

- UCS and STS improved with increasing AA content and reached their peak at 20% RHA and 15% AA. The highest UCS recorded was 10,234.47 kPa after 28 days of curing. However, beyond 20% RHA, a decline in strength was observed due to excessive silica content, which hindered optimal geopolymerization.
- The LCA results reveal that the alkali-activated mix (20%RHA+80%CS+3%AA) has significantly lower environmental impacts than the cement-based mix (20%RHA+80%CS+5%C), reducing CO₂ emissions, acidification, eutrophication, and energy consumption per MPa of strength. Although ecotoxicity is slightly higher due to metallic content, overall, the mix offers a sustainable and eco-efficient alternative for pavement subgrade applications.
- FE-SEM and EDX results confirmed the formation of dense geopolymer gels (N-A-S-H and C-A-S-H), contributing to the composite's strength. XRD analysis revealed the presence of strength-enhancing compounds, including calcium silicate hydrates and aluminosilicate gels, validating the chemical stability of the developed material.
- The mix containing 20% RHA, 80% CS, and 3% AA (M1A2 from Mix-1) was identified as the most cost-effective formulation due to its lower alkali activator dosage while still meeting the IRC-specified strength requirements. In contrast, the optimized mix with 20% RHA, 80% CS, and 15% AA (M5A2 from Mix-5) exhibited superior strength performance. Together, these findings demonstrate a balance between mechanical efficiency and economic sustainability, highlighting the potential of CS-RHA-AA composites as eco-friendly and high-performance materials for pavement subgrade applications.
- The multi-linear regression model (MLRA) exhibited a strong correlation between the predicted and experimental values for MDD ($R^2 = 0.95$ training, $R^2 = 0.91$ testing), OMC ($R^2 = 0.95$ training, $R^2 = 0.97$ testing), UCS ($R^2 = 0.84$ training, $R^2 = 0.80$ testing), and STS ($R^2 = 0.83$ training, $R^2 = 0.79$ testing), confirming the reliability and robustness of the proposed predictive model.
- The use of CS and RHA in pavement construction offers an environmentally friendly alternative to traditional materials like cement and lime. This approach not only reduces industrial waste disposal issues but also minimizes CO₂ emissions, supporting sustainable construction practices.

In conclusion, the developed geopolymer composite, incorporating CS, RHA, and AA, exhibits promising mechanical

properties suitable for subgrade and sub-base layers in pavement construction. Further research is recommended to evaluate its long-term durability under real-world traffic and environmental conditions.

Recommendations and field application challenges

Based on the findings, alkali activator (AA) can be effectively utilized as a fully replacement for cement. The effective utilization of AA is up to 15%, with the optimum proportions of 80% copper slag and 20% for rice husk ash. Further studies are required to evaluate long-term performance under real environmental conditions, including durability (freeze-thaw and wet-dry cycles), temperature variations (high temperature), and the effect of chemical exposure. Another critical concern is the variability in the chemical and physical composition of industrial by-products such as copper slag and rice husk ash, which can influence the reactivity and consistency of the final product. Pilot-scale testing is also recommended prior to implementation to validate the practical applicability of the CS-RHA-AA composite mix for subgrade construction in flexible pavements.

To mitigate these challenges, it is recommended to develop standardized preprocessing and characterization protocols for raw materials, adopt pre-mixed activator solutions to ensure safe handling, and explore ambient-curable mix formulations suitable for field conditions.

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s10163-026-02579-3>.

Declarations

Competing interests We know of no conflicts of interest associated with this publication, and there has been no significant financial support for this work that could have influenced its outcome. As corresponding author, I confirm that the manuscript has been read and approved for submission by all the named authors.

References

1. Singh J, Laurenti R, Sinha R, Frostell B (2014) Progress and challenges to the global waste management system. *Waste Manag Res* 32:800–812. <https://doi.org/10.1177/0734242X14537868>
2. Dixit A, Kumar Das S (2025) Mechanical, microstructural, and durability assessment of cement slurry waste and coal combustion ash mixture as a sustainable subgrade construction material: Experimental and mechanistic modeling approach. *Constr Build Mater* 458:139550. <https://doi.org/10.1016/j.conbuildmat.2024.139550>
3. Gupta D, Kumar A (2017) Performance evaluation of cement-stabilized pond ash-rice husk ash-clay mixture as a highway

- construction material. *J Rock Mech Geotech Eng* 9:159–169. <https://doi.org/10.1016/j.jrmge.2016.05.010>
4. Sharma K, Kumar A (2021) Influence of rice husk ash, lime and cement on compaction and strength properties of copper slag. *Transp Geotech* 27:100464. <https://doi.org/10.1016/j.trgeo.2020.100464>
 5. Norouzi M, Châfer M, Cabeza LF et al (2021) Circular economy in the building and construction sector: A scientific evolution analysis. *J Build Eng* 44:1–18. <https://doi.org/10.1016/j.jobe.2021.102704>
 6. Ghafoor S, Shooshtarian S, Udawatta N et al (2024) Cost factors affecting the utilisation of secondary materials in the construction sector: A systematic literature review. *Resour Conserv Recycl Adv* 23:1–14. <https://doi.org/10.1016/j.rcradv.2024.200230>
 7. Ilkentalpar S, Örklemmez E, Durak U et al (2025) Evaluation of diatomite substitute with thermal power plant waste fly ash in sustainable geopolymer through life cycle assessment. *J Mater Cycles Waste Manag* 27:1418–1435. <https://doi.org/10.1007/s10163-025-02184-w>
 8. Huntzinger DN, Eatmon TD (2009) A life-cycle assessment of Portland cement manufacturing: comparing the traditional process with alternative technologies. *J Clean Prod* 17:668–675. <https://doi.org/10.1016/j.jclepro.2008.04.007>
 9. Alrawashdeh AA, Zamorano M, Alshaaer M, Martín-Morales M (2024) Promoting the use of geopolymer and alkali-activated materials through the identification of critical factors and strategies. *J Build Eng* 98:1–17. <https://doi.org/10.1016/j.jobe.2024.111078>
 10. Nasir M, Mahmood AH, Bahraq AA (2024) History, recent progress, and future challenges of alkali-activated binders-An overview. *Constr Build Mater* 426:1–20. <https://doi.org/10.1016/j.conbuildmat.2024.136141>
 11. Kumar Das S, Adediran A, Rodrigue Kaze C et al (2022) Production, characteristics, and utilization of rice husk ash in alkali activated materials: An overview of fresh and hardened state properties. *Constr Build Mater* 345:128341. <https://doi.org/10.1016/j.conbuildmat.2022.128341>
 12. Duxson P, Provis JL, Lukey GC, van Deventer JSJ (2007) The role of inorganic polymer technology in the development of green concrete. *Cem Concr Res* 37:1590–1597. <https://doi.org/10.1016/j.cemconres.2007.08.018>
 13. Singh R, Rao B, Asolekar SR (2025) Geopolymer-based solidification and stabilization for environmentally sound disposal of asbestos-containing waste. *J Mater Cycles Waste Manag* 27:75–90. <https://doi.org/10.1007/s10163-024-02076-5>
 14. Li X, Li Z (2025) Study on clinker ash recycling in geopolymer materials. *J Mater Cycles Waste Manag* 27:289–304. <https://doi.org/10.1007/s10163-024-02112-4>
 15. Hoai TT, Mukunoki T, Shida K et al (2025) Adsorption, diffusion, and hydraulic performance of geopolymer from fly ash waste-bentonite clay in ammonium solution. *J Mater Cycles Waste Manag* 27:2467–2481. <https://doi.org/10.1007/s10163-025-02253-0>
 16. Ulandari O, Sujoto VSH, Astuti W et al (2025) Enhancement of compressive strength in geopolymers: a study on fly ash characteristics and solid activation using dry method. *J Mater Cycles Waste Manag* 27:2303–2320. <https://doi.org/10.1007/s10163-025-02233-4>
 17. Kumar MGR, Parvathikumar G, Arunkumar GE, Rajeshkumar G (2025) Investigating the properties of agro-waste fiber reinforced low calcium fly ash geopolymer composites: the effect of alkali treatment and nano-clay addition. *J Mater Cycles Waste Manag* 27:984–995. <https://doi.org/10.1007/s10163-024-02152-w>
 18. Singh P, Kapoor K (2024) Assessing the impact of recycled mortar powder on the strength, durability and microstructural characteristics of geopolymer concrete. *J Mater Cycles Waste Manag* 26:2741–2753. <https://doi.org/10.1007/s10163-024-01999-3>
 19. Ponomar V, Yliniemi J, Adesanya E et al (2022) An overview of the utilisation of Fe-rich residues in alkali-activated binders: Mechanical properties and state of iron. *J Clean Prod* 330:1–20. <https://doi.org/10.1016/j.jclepro.2021.129900>
 20. El Fadili H, Ait-Khouia Y, Ouffa N et al (2025) Comprehensive review on alkaline dissolution of aluminosilicates and its role in the mechanisms and properties of geopolymers and alkali-activated materials. *Case Stud Constr Mater* 23:1–28. <https://doi.org/10.1016/j.cscm.2025.e05330>
 21. Alloul A, Amar M, Benzerzour M, Abriak NE (2025) High strength, durability, and microstructure of geopolymer concrete with flash-calcined soils cured under ambient conditions. *J Mater Cycles Waste Manag* 27:1854–1872. <https://doi.org/10.1007/s10163-025-02209-4>
 22. Rawat MS, Kaloni S, Tipu RK, Bhandari P (2025) Sustainability and economic evaluation of recycled aggregate geopolymer concrete using fly ash and slag. *J Mater Cycles Waste Manag* 1–15. <https://doi.org/10.1007/s10163-025-02373-7>
 23. Pesaramelli MR, Nayaka R, Siva Kumar MVN (2025) Performance of geopolymer concrete for paver blocks in light traffic applications by incorporating granite waste as fine aggregate replacement. *J Mater Cycles Waste Manag* 27:2352–2369. <https://doi.org/10.1007/s10163-025-02237-0>
 24. Montazeri P, Bamshad O, Aghililoft M, Singh P (2025) Mechanical and durability-based life cycle assessment of rice husk ash containing concrete. *Case Stud Constr Mater* 23:1–31. <https://doi.org/10.1016/j.cscm.2025.e05241>
 25. Gorai B, Jana RK (2003) Characteristics and utilisation of copper slag-a review. *Resour Conserv Recycl* 39:299–313. [https://doi.org/10.1016/S0921-3449\(02\)00171-4](https://doi.org/10.1016/S0921-3449(02)00171-4)
 26. Phiri TC, Singh P, Nikoloski AN (2022) The potential for copper slag waste as a resource for a circular economy: A review-Part I. *Min Eng* 180:1–14. <https://doi.org/10.1016/j.mineng.2022.107474>
 27. Sharma K, Kumar A (2022) Investigation of compaction, specific gravity, unconfined compressive strength and CBR of a composite having copper slag and rice husk ash mixed using an alkali activator. *Innov Infrastruct Solut* 7:1–23. <https://doi.org/10.1007/s41062-022-00783-2>
 28. Singh J, Singh SP (2019) Development of Alkali-activated Cementitious Material using Copper Slag. *Constr Build Mater* 211:73–79. <https://doi.org/10.1016/j.conbuildmat.2019.03.233>
 29. Moura WA, Gonçalves JP, Lima MBL (2007) Copper slag waste as a supplementary cementing material to concrete. *J Mater Sci* 42:2226–2230. <https://doi.org/10.1007/s10853-006-0997-4>
 30. Afshoon I, Miri M, Mousavi SR (2023) Comprehensive experimental and numerical modeling of strength parameters of eco-friendly steel fiber reinforced SCC containing coarse copper slag aggregates. *Constr Build Mater* 367:1–28. <https://doi.org/10.1016/j.conbuildmat.2023.130304>
 31. Al-Jabri KS, Hisada M, Al-Oraimi SK, Al-Saidy AH (2009) Copper slag as sand replacement for high performance concrete. *Cem Concr Compos* 31:483–488. <https://doi.org/10.1016/j.cemconcomp.2009.04.007>
 32. Hamada HM, Al-Attar A, Beddu S et al (2025) Impact of rice husk ash on geopolymer concrete: A literature review and future directions. *Case Stud Constr Mater* 22:1–22. <https://doi.org/10.1016/j.cscm.2025.e04476>
 33. Celik F, Canakci H (2015) An investigation of rheological properties of cement-based grout mixed with rice husk ash (RHA). *Constr Build Mater* 91:187–194. <https://doi.org/10.1016/j.conbuildmat.2015.05.025>
 34. Rajak AK, Mahato DL, Dalal S et al (2026) Development, assessment, and efficiency of rice husk ash (RHA) infused by eggshell as a hybrid catalyst towards biodiesel synthesis from iluppai

- ennai oil. *Biomass Bioenergy* 204:1–20. <https://doi.org/10.1016/j.biombioe.2025.108347>
35. ASTM C618-22 (2017) Standard Specification for Coal Fly Ash and Raw or Calcined. Natural Pozzolan for Use in Concrete
 36. Yoobanpot N, Jamsawang P (2014) Effect of cement replacement by rice husk ash on soft soil stabilization. *Kasetsart J - Nat Sci* 48:323–332
 37. Anto G, Athira K, Nair NA et al (2022) Mechanical properties and durability of ternary blended cement paste containing rice husk ash and nano silica. *Constr Build Mater* 342:1–9. <https://doi.org/10.1016/j.conbuildmat.2022.127732>
 38. Kumar A, Gupta D (2016) Behavior of cement-stabilized fiber-reinforced pond ash, rice husk ash-soil mixtures. *Geotext Geomembranes* 44:466–474. <https://doi.org/10.1016/j.geotexmem.2015.07.010>
 39. Sharma K, Kumar A (2023) Performance evaluation of lime-cement-copper slag-rice husk ash composite as a pavement construction material. *Int J Pavement Eng* 24:1–21. <https://doi.org/10.1080/10298436.2022.2121833>
 40. Shekhawat P, Sharma G, Singh RM (2020) Potential Application of Heat Cured Eggshell Powder and Flyash-Based Geopolymer in Pavement Construction. *Int J Geosynth Gr Eng* 6:1–17. <https://doi.org/10.1007/s40891-020-00213-2>
 41. Tekin I (2016) Properties of NaOH activated geopolymer with marble, travertine and volcanic tuff wastes. *Constr Build Mater* 127:607–617. <https://doi.org/10.1016/j.conbuildmat.2016.10.038>
 42. Sharma K, Kumar A (2020) Utilization of industrial waste-based geopolymers as a soil stabilizer-a review. *Innov Infrastruct Solut* 5:1–20. <https://doi.org/10.1007/s41062-020-00350-7>
 43. Wang L, Gao Z, Zhang W et al (2025) Effect of activator modulus on the interfacial bonding behavior between high ductility geopolymer composites and concrete substrate. *Constr Build Mater* 487:1–14. <https://doi.org/10.1016/j.conbuildmat.2025.142149>
 44. Yan Z, Sun Z, Yang J et al (2021) Mechanical performance and reaction mechanism of copper slag activated with sodium silicate or sodium hydroxide. *Constr Build Mater* 266:120900. <https://doi.org/10.1016/j.conbuildmat.2020.120900>
 45. Shahu JT, Patel S, Senapati A (2013) Engineering Properties of Copper Slag–Fly Ash–Dolime Mix and Its Utilization in the Base Course of Flexible Pavements. *J Mater Civ Eng* 25:1871–1879. [https://doi.org/10.1061/\(ASCE\)MT.1943-5533.0000756](https://doi.org/10.1061/(ASCE)MT.1943-5533.0000756)
 46. C.Lavanya A, Sreerama Rao NDK (2013) Study on Coefficient of Permeability of Copper slag when admixed with Lime and Cement. *J Mech Civ Eng* 7:19–25. <https://doi.org/10.9790/1684-0761925>
 47. Canakci H, Aziz A, Celik F (2015) Soil stabilization of clay with lignin, rice husk powder and ash. *Geomech Eng* 8:67–79. <https://doi.org/10.12989/gae.2015.8.1.067>
 48. ASTM D 854–02 (2006) Standard Test Methods for of Soil Specific Gravity Solids by Water Pycnometer. ASTM Stand 24:120432
 49. ASTM D1557-07 (2002) Standard Test Methods for Laboratory Compaction Characteristics of Soil Using Modified Effort. 1–13
 50. Anagnos JN, Kennedy TW, Hudson WR (1970) Evaluation and Prediction of Tensile. Properties of Cement-Treated Materials
 51. ASTM D2166–00 (2000) Standard Test Method for Unconfined Compressive Strength of Cohesive Soil. ASTM Int West Conshohocken, PA D2166-00:1–6
 52. ASTM C496/C496M-04 (2006) Standard Test Method for Splitting Tensile Strength of Cylindrical Concrete Specimens. 1–5
 53. ISO 14040 (2009) Environmental assessment - Life cycle assessment - Principles and framework. *Int Stand Organ* 1997:1–20
 54. 14044:2006 I (2006) Environmental Management. Life Cycle Assessment. Requirements and Guidelines (Edition 1, 2006). Int Stand confirmed 2006:46
 55. Bare J (2011) TRACI 2.0: The tool for the reduction and assessment of chemical and other environmental impacts 2.0. *Clean Technol Environ Policy* 13:687–696. <https://doi.org/10.1007/s10098-010-0338-9>
 56. Mishra A, Das SK, Reddy KR (2023) Life cycle assessment of processing alternate sands for sustainable construction: Coal mine overburden sand versus manufactured sand. *Resour Policy* 86:1–15. <https://doi.org/10.1016/j.resourpol.2023.104140>
 57. Wang R, Shi Q, Li Y et al (2021) A critical review on the use of copper slag (CS) as a substitute constituent in concrete. *Constr Build Mater* 292:1–21. <https://doi.org/10.1016/j.conbuildmat.2021.123371>
 58. Mehta PK, Monteiro PJM (2001) Concrete-Microstructure, Properties, and Materials
 59. Abdullah HH, Shahin MA, Sarker P (2019) Use of Fly-Ash Geopolymer Incorporating Ground Granulated Slag for Stabilisation of Kaolin Clay Cured at Ambient Temperature. *Geotech Geol Eng* 37:721–740. <https://doi.org/10.1007/s10706-018-0644-2>
 60. Poltue T, Suddepong A, Horpibulsuk S et al (2020) Strength development of recycled concrete aggregate stabilized with fly ash-rice husk ash based geopolymer as pavement base material. *Road Mater Pavement Des* 21:2344–2355. <https://doi.org/10.1080/14680629.2019.1593884>
 61. Alam S, Das SK, Rao BH (2019) Strength and durability characteristic of alkali activated GGBS stabilized red mud as geomaterial. *Constr Build Mater* 211:1–11. <https://doi.org/10.1016/j.conbuildmat.2019.03.261>
 62. IRC:37 (2018) Guidelines for the design of flexible pavements. Indian Road Congr 1–96
 63. IRC:SP-72 (2015) Guidelines for the Design of Flexible Pavements for Low Volume Rural Roads. Indian Road Congr 1–48
 64. Dixit A, Jain S, Kumar S et al (2024) A systematic characterization of the mixture of red mud and bottom ash as a geomaterial: an efficient utilization in subgrade pavement. *J Mater Cycles Waste Manag*. <https://doi.org/10.1007/s10163-024-01955-1>
 65. Davidovits J (1994) Geopolymers: Man-Made Rock Geosynthesis and the Resulting Development of Very Early High Strength Cement. *Mater Education* 16:1–25
 66. Provis JL, Bernal SA (2014) Geopolymers and related alkali-activated materials. *Annu Rev Mater Res* 44:299–327. <https://doi.org/10.1146/annurev-matsci-070813-113515>
 67. Xu H, Van Deventer JSJ (2000) The geopolymerisation of aluminosilicate minerals. *Int J Min Process* 59:247–266. [https://doi.org/10.1016/S0301-7516\(99\)00074-5](https://doi.org/10.1016/S0301-7516(99)00074-5)
 68. Palomo A, Grutzeck MW, Blanco MT (1999) Alkali-activated fly ashes: A cement for the future. *Cem Concr Res* 29:1323–1329. [https://doi.org/10.1016/S0008-8846\(98\)00243-9](https://doi.org/10.1016/S0008-8846(98)00243-9)

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.